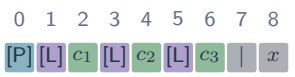
$$U = H[P,:], \quad A = \mathrm{ReLU}(UW_1^\mathsf{T} + b_1), \quad s = \mathrm{squeeze}(AW_2^\mathsf{T} + b_2)$$

1. Token layout

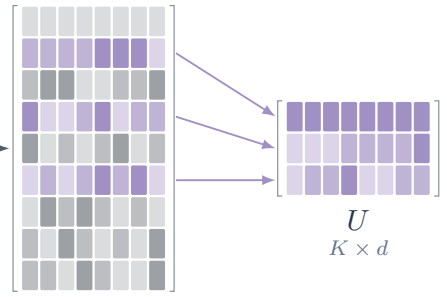


x
T integer token IDs

2. GLiNER2

DeBERTa-v3-base
12 bidirectional blocks; d=768
legacy span classification

3. Contextual states and indices



H
$T \times d$

U
$K \times d$

Read [L], skip [P]
no CLS pooling
joint bidirectional encoding

4. Shared d-to-2d expansion and ReLU



U
$K \times d$

W_1^T
$d \times m$

A
$K \times m$

5. Scalar contraction, squeeze and activation



A
$K \times m$

W_2^T
$m \times 1$

v
$K \times 1$

s
K

p
K

Axes $d = 768$, $m = 1536$; Single-question illustration; batch omitted. Tile counts encode axes, not literal sizes.

Objects Gray: input/context; violet: hidden states; orange: parameter matrices; red: logits/probabilities. Positions, code IDs and owner maps are integers, not scores. Tiles show roles, not measured magnitudes.

Mechanism Gather actual [L] markers, skipping [P]; shared d-to-2d ReLU-to-1 MLP. The processor does not insert a CLS readout for this path.

Classification uses sum-reduced BCE even when single-label inference selects softmax. The span/count heads are not on this classification path.